Kinwise.

Family medicine-review co-pilot · private, on-device AI

UnivaBio 2026 · AI for Human Health · Fictional demo data

Working prototype · runs in the browser, no backend

Helps families caring for an older adult on many medicines notice what changes after a medicine change, and bring a clear question to the visit.

Live demo (no install): albertzorzolan.github.io/UnivaBioBabson → Try as Alex **Demo video:** on the Devpost page

Code: github.com/albertzorzolan/UnivaBioBabson/tree/medwatch-v2

THE PROBLEM

43.0%
of US adults 65+ took **5 or more prescription drugs** in a 30-day period (2017–Mar 2020).
NCHS, Health, United States

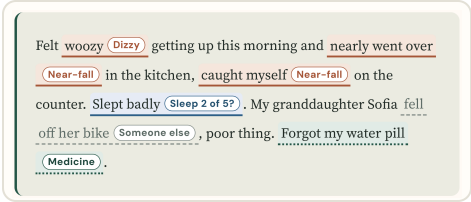
37%
of adverse drug events in primary care whose harm could have been reduced were attributed to **the patient not telling the doctor**.
Gandhi et al., NEJM 2003

More than 1 in 4
older adults **fall each year**, but **less than half tell their doctor**.
CDC, Facts About Older Adult Falls

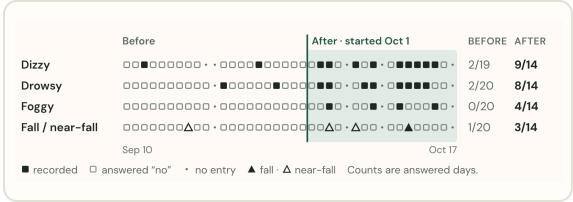
What's different. Diary apps (MyTherapy, CareClinic) record and chart symptoms but do not document a before/after count against a medicine's start date; clinician tools (MedStopper, MedSafer) apply Beers/STOPP-type rules without the family's home observations. Kinwise does both, on the device. (Vendor pages, Oct 2026; "not documented" is not proof of absence.)



Alex: before vs after



Margaret (patient): her words, read on-device



Visit brief strip · full sample in the brief PDF

WHO IT IS FOR, AND WHERE

The **family caregiver** (Alex) and the **older adult** (Margaret, 78, 8 medicines), who checks in on a large-text screen; the brief is for her **doctor or pharmacist**.

Designed for use before a Medicare **Annual Wellness Visit** or a pharmacist's **Comprehensive Medication Review**, which may be done with the caregiver if she cannot take part because of cognitive impairment (42 CFR 423.153(d)(1)(vii)(B)(2)).

> **Why it could help:** in EMPOWER (JAMA Intern Med 2014; cluster RCT, 303 long-term benzodiazepine users aged 65–95), **62%** of those given a brochure raised stopping with their doctor or pharmacist, and **27% vs 5%** (brochure vs usual care) had discontinued at 6 months. A reason to test Kinwise, not proof it works; Kinwise only prompts the conversation.

WHAT IT DOES

1. **Check in, in her own words.** On-device NLP turns "felt woozy" into answers she confirms.
2. **Review her medicines** against AGS Beers 2023, anticholinergic burden, fall-risk drugs and cascades; each finding is cited and ends as a question.
3. **See what changed.** Dizziness on **2 of 19** answered days before lorazepam, **9 of 14** after (Fisher's exact test).
4. **Bring one reviewed page to the visit,** plus a FHIR R4 export.

HOW IT WORKS

Cited KB → Engines → LLM rewords → Code checks → Human review

Deterministic TypeScript engines over a cited knowledge base (427 medicines, 89 Beers rules, 26 cascades) compute every fact. An optional on-device LLM (Qwen2.5, WebGPU) only rewords them, and code drops any sentence that fails its number, name and wording checks. No account, no server.

EVIDENCE & RIGOR

- **Clinical NLP:** micro-F1 **94.2%** (95% CI 90.7–97.1%) on 70 held-out stories never used for tuning; lexicon-only baseline 70.3%.
- **Medication review: 57 of 57** synthetic cases pass (a written spec, not clinical validity); ablated engines: 33–54 of 57.
- **Claim checker red team** (AI-written, not clinician-reviewed): **72 of 163** harmful drafts caught as submitted, 163 of 163 after fixes tuned on that set; new paraphrases not used for tuning: 12 of 20 caught; 0 of 87 valid sentences rejected.
- **Timeline false alarms** (1,000 simulations, no real change): moderate-or-strong signal on 11.2% of changes; 85.8% if each answer repeats the day before with probability 0.8 (assumed).
- **On-device AI:** 0 external requests in a recorded run; at least 6 flawed sentences got through; the template stays the default.
- **Software:** 1,453 unit tests pass · 21 of 21 end-to-end checks pass · 46 axe scans, 0 violations.

WORKING

- Public live demo, no install
- NLP check-in · medicine matching
- Cited review · before/after · Ask Kinwise
- Reviewed brief · print · FHIR R4 export

PARTIAL

- On-device LLM: optional, WebGPU only
- KB: 427 medicines, no clinician review
- No dose or allergy checks · English only
- One device, no sync · no FHIR receiver

PLANNED (NOT BUILT)

- Pharmacist KB review · validation pilot
- Encrypted family sharing and storage
- Pharmacy, portal import · voice · Spanish

Live demo: albertzorzolan.github.io/UnivaBioBabson
Video: on the Devpost page
Code: github.com/albertzorzolan/UnivaBioBabson/tree/medwatch-v2
Team: Albert Zorzolan and Neil Shah, Babson College

Hackathon prototype: not FDA reviewed or cleared, not clinically validated, no HIPAA claim. Designed never to say a medicine caused a symptom or to advise a change; the clinician decides. Emergency: call 911.